

1. Administrative & Study Identification

Full Study Title: A Phase 1 Open-label, Multicenter Study of MK-6837 as Monotherapy and Combination Therapy in Participants With Advanced/Metastatic Solid Tumors **Short Title/Acronym:** MK-6837-001 **Protocol Number:** MK-6837-001 (Other ID: 6837-001) **NCT Number:** NCT06460961 **Posting ID:** 124368695430445934 **Trial Status:** Recruiting **Sponsor & Company Name:** Merck **Investigational Product:** MK-6837 (monotherapy) and MK-6837 in combination with Pembrolizumab (MK-3475; Trade Name: Keytruda; ATC Code: L01FF02) **Phase of Development:** Phase 1 **Medical Monitor:** Merck Clinical Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, pharmacokinetics (PK), and preliminary efficacy of MK-6837, administered as a monotherapy and in combination with pembrolizumab (MK-3475), in participants with histologically or cytologically confirmed advanced/metastatic solid tumors that have not responded to conventional therapy. There will not be any hypothesis testing in the study. **Secondary Objectives:** To evaluate the pharmacokinetics (PK) of MK-6837, specifically focusing on maximum concentration, minimum concentration, and area under the concentration-time curve.

2.2 Background & Rationale To provide a novel therapeutic approach by potentially enhancing the immune response against tumors in patients with advanced/metastatic solid tumors that have not responded to conventional therapy. MK-6837 is designed to work in tandem with pembrolizumab to possibly boost the immune system's ability to fight cancer more effectively, exploring beyond standard immune checkpoint inhibitor monotherapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm/Dose-escalation cohorts (No placebo) **Blinding:** Open-label **Randomization:** Non-Randomized (Parallel Assignment)

3.2 Planned Enrollment & Duration Target \$N\$: 168 participants (Estimated) **Number of Sites:** Multicenter **Estimated Study Start:** 14-Jul-2024 **Estimated Primary Completion:** 13-Jul-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed solid tumor by pathology report that is advanced or metastatic. 2. Human Immunodeficiency Virus (HIV)-infected participants must have well controlled HIV on Antiretroviral Therapy (ART). 3. Participants who are Hepatitis B Surface Antigen (HBsAg) positive are eligible if they have received Hepatitis B Virus (HBV) antiviral therapy for at least 4 weeks, and have undetectable HBV viral load before allocation. 4. Participants with history of Hepatitis C Virus (HCV) infection are eligible if HCV viral load is undetectable at screening.

4.2 Exclusion Criteria 1. Has not recovered to Common Terminology Criteria for Adverse Events (CTCAE) Grade 1 or better from any Adverse Events (AEs) that were due to cancer therapeutics administered more than 4 weeks earlier. 2. History of a second malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 2 years. 3. Has clinically significant cardiovascular disease. 4. HIV-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 5. Received prior systemic anticancer therapy including investigational agents within 4 weeks before the first dose of study intervention. 6. Has received any prior immunotherapy and was discontinued from that treatment due to a Grade 3 or higher immune-related AE (except endocrine disorders that can be treated with replacement therapy) or was discontinued from that treatment due to Grade 2 myocarditis or recurrent Grade 2 pneumonitis. 7. Received prior radiotherapy within 2 weeks of start of study intervention, or has radiation-related toxicities, requiring corticosteroids. 8. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. Administration of killed vaccines is allowed. 9. Diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy or any other form of immunosuppressive therapy within 7 days before the first dose of study intervention. 10. Known additional malignancy that is progressing or has required active treatment within the past 2 years. 11. Known active Central Nervous System (CNS) metastases and/or carcinomatous meningitis. 12. Active autoimmune disease that has required systemic treatment in the past 2 years except replacement therapy. 13. History of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 14. Active infection requiring systemic therapy. 15. History of allogeneic tissue/solid organ transplant. 16. Participants who have not adequately recovered from major surgery or have ongoing surgical complications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Escalating doses of MK-6837 via IV infusion once every 3 weeks (Q3W, Day 1 of 21-day cycle), evaluated both alone and PLUS 200 mg of pembrolizumab (Keytruda, ATC: L01FF02) via IV infusion Q3W. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Until progressive disease or discontinuation (up to 35 administrations or approximately 2 years for the pembrolizumab combination).

5.2 Concomitant & Prohibited Therapy Permitted: Routine supportive care, protocol-approved rescue medications, and killed vaccines. **Prohibited:** Administration of live or live-attenuated vaccines within 30 days before the first dose; systemic steroid therapy or other forms of immunosuppressive therapy within 7 days before the first dose; prior radiotherapy within 2 weeks requiring corticosteroids.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	-28 to 0	Informed Consent, Medical History, Baseline Scans, Baseline Labs
Cycle 1 (Day 1)	Day 0	First IV infusion of MK-6837 + Pembrolizumab, Safety Labs, DLT Observation window starts
Treatment Cycles	Every 21 Days (Q3W)	Safety Labs, Efficacy Assessment, Continuous AE Monitoring
End of Study	Up to ~59 Months	Final Safety/Efficacy Scans, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Number of participants who experience one or more dose-limiting toxicities (DLTs):** Evaluated during the first 21 days. DLTs include events unless clearly due to underlying disease/extraneous causes: Grade 4 neutropenia lasting >7 days; Grade 3+ thrombocytopenia associated with clinically significant bleeding; all Grade 3+ nonhematologic toxicities (with exceptions); drug induced liver injury; febrile neutropenia Grade 3 or 4; prolonged treatment delay (>2 weeks); intervention discontinuation during first 21 days due to toxicity; Grade 5 toxicity.
- 2. Number of participants who experience one or more adverse events (AEs):** An AE is any untoward medical occurrence temporally associated with the use of study intervention, regardless of relatedness.
- 3. Number of participants who discontinue study intervention due to an AE.**

7.2 Secondary Endpoints

- 1. Maximum concentration (C_{max}) of MK-6837 in plasma.**
- 2. Minimum concentration (C_{trough}) of MK-6837 in plasma.**
- 3. Area Under the Concentration-Time Curve (AUC) of MK-6837 in plasma.**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: An AE is defined as any untoward medical occurrence temporally associated with the use of the study intervention, whether or not considered related to it.

Serious Adverse Events (SAEs): Continuously tracked; intervention-related toxicities causing treatment delays >2 weeks or discontinuation in the first 21 days are formally flagged as DLTs. **Data Monitoring Committee (DMC):** Internal safety review and dose-escalation committee to monitor DLTs before stepping up dosages.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for safety/DLT endpoints (all participants who receive at least 1 dose of study intervention).

Sample Size Justification: Target enrollment is $N = 168$, evaluated under standard Phase 1 dose-escalation methodologies.

Hypothesis Testing: There will not be any formal hypothesis testing in this study. Descriptive statistics will be utilized.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: High-frequency site monitoring during the Cycle 1 (21-day) DLT observation window to dictate dose escalation.

Audit Trail: Sponsor eCRF locking and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: MK-6837-001 **Registry Number:** NCT06460961 **Posting ID:** 124368695430445934 **Trial Status:** Recruiting **Sponsor/Lead**

Organization: Merck **Study Title:** A Study of MK-6837 as a Monotherapy and Combination Therapy in Participants With Advanced/Metastatic Solid Tumors (MK-6837-001) **Official**

Regulatory Title: A Phase 1 Open-label, Multicenter Study of MK-6837 as Monotherapy and Combination Therapy in Participants With Advanced/Metastatic Solid Tumors

12. Clinical Framework (Oncology Specific)

Primary Condition / Disease Area: Neoplasm Metastasis **UMLS /**

Ontology Details: Neoplasm Metastasis (Semantic Type: Neoplastic Process; MeSH: D009362; SNOMED CT: 128469006) **Diagnosis**

Threshold: Histologically or cytologically confirmed solid tumor

that has not responded to conventional therapy. **Medical Methodology:** Investigational Monotherapy and Checkpoint Inhibitor Combination Therapy. **Optimization Target:** Maximum Tolerated Dose (MTD) / Safety, Tolerability, and Pharmacokinetics evaluation.

13. Study Procedures & Methodology

Management Pathway: Phase 1 dose-escalation oncology clinical pathway **Education Protocol (HCP):** Dose-escalation safety monitoring and rigorous DLT criteria training **Education Protocol (Patient):** Informed consent, infusion scheduling, and strict AE reporting instructions **Quality Audit Mechanism:** Continuous clinical safety review and formal Dose-Escalation Committee evaluations

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 59 months **Onsite Visit Cadence:** Every 3 weeks (Q3W) for IV infusion and assessments **Remote Monitoring Frequency:** Routine safety checks between dosing cycles as required **Checklist Review Interval:** Every 21 days (end of each cycle)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				